Software Requirements Specification (SRS)

Public Transport Management System (PTMS)

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the Public Transport Management System (PTMS). The system will enhance efficiency, transparency, and user experience in the public transportation sector with respect to real-time monitoring, digital services, and data-driven decision making.

1.2 Scope

PTMS is a web and mobile based system that is used in the operations of the public transport. It serves various stakeholders such as passengers, drivers, conductors, bus owners and transport authorities. Key features include

- Live bus tracking
- Crowd monitoring and prediction
- Digital ticketing (optional)
- Revenue and maintenance management
- Welfare systems for staff
- Demand forecasting and route optimization
- Customer complaint management

1.3 Definitions, Acronyms, and Abbreviations

- PTMS – Public Transport Management System
- GPS – Global Positioning System
- ETA – Estimated Time of Arrival
- RBAC – Role-Based Access Control
- API – Application Programming Interface

1.4 References

- IEEE 830-1998 Software Requirements Specification Standard
- Intelligent Transport Systems (ITS) Guidelines

1.5 Overview

This document describes the overall system, user roles, functional requirements, non-functional requirements, and system constraints for PTMS.

2. Overall Description

2.1 Product Perspective

PTMS will integrate GPS devices, mobile applications, cloud servers, and analytics engines. It can be integrated with existing transport authority systems and payment gateways.

2.2 User Characteristics

Passenger :

- Live bus tracking
- Customers Complain
- Live crowd Display
- Crowd prediction
- Digital Ticket (optional)

Driver and : Conductor

- Driver and Conductor welfare system
- Issue Tickets and verify tickets
- Emergency alert
- Timetable / duty roster
- Rotational assignment (shift-based, route-based)

Bus owners:

- Daily Revenue generator
- Tax & cost calculation
- Bus maintain record maintaining and Welfare system

Authorities:

- Traffic and demand forcast
- Dynamic bus route allocation
- Gather customer complains

2.3 Constraints

- Must operate under local transport regulations.
- Internet connectivity may be unstable in some areas.
- System must support high concurrent users during peak hours.

3. Functional Requirements

3.1 Passenger Module

- FR1.1: The system shall provide live bus tracking using GPS.
- FR1.2: The system shall display estimated arrival times (ETA).
- FR1.3: The system shall display live crowd levels in buses.
- FR1.4: The system shall predict crowd levels based on historical data.
- FR1.5: Passengers shall be able to submit complaints and feedback.
- FR1.6: The system may support digital ticket booking and validation.

3.2 Driver & Conductor Module

- FR2.1: The system shall allow drivers and conductors to log in securely.
- FR2.2: The system shall record duty hours and trip assignments.
- FR2.3: The system shall provide access to welfare-related benefits.
- FR2.4: Emergency alerts shall be available for drivers and conductor
- FR2.5: The system shall integrate with the conductor's ticket issuing machine to capture real-time GPS location data and transmit it to the central system.
- FR2.6: The conductor shall be able to verify passenger details and issue digital or printed tickets using the ticket issuing machine
- FR2.7: The system shall automatically record issued ticket details, including route, fare, time, and GPS location.

3.3 Bus Owner Module

- FR3.1: The system shall generate daily revenue reports.
- FR3.2: The system shall calculate taxes and operational costs.
- FR3.3: The system shall maintain bus maintenance and service records.
- FR3.4: The system shall manage welfare schemes for staff.

3.4 Authority Module

- FR4.1: The system shall provide traffic and passenger demand forecasts.
- FR4.2: The system shall support dynamic bus route allocation.
- FR4.3: Authorities shall be able to view and manage passenger complaints.
- FR4.4: The system shall generate analytics reports for planning.

4. Non-Functional Requirements

4.1 Performance Requirements

- System response time should be less than 3 seconds.
- Must support at least 5,000 concurrent users.

4.2 Security Requirements

- User data must be encrypted using industry standards.
- Implement role-based access control (RBAC).
- Secure authentication for all user roles.

4.3 Availability & Reliability

- System uptime should be at least 99.9%.
- Automatic daily data backups must be maintained.

4.4 Usability

- User-friendly interfaces for non-technical users.
- Support for multiple local languages.

4.5 Scalability

- System should scale to support additional cities and routes.

5. System Requirements

5.1 Hardware Requirements

- Server: Minimum 8-core CPU, 32GB RAM, 1TB SSD
- GPS devices for buses
- Smartphones for drivers/conductors

5.2 Software Requirements

- Operating System: Linux / Windows Server
- Database : PostgreSQL
- Frontend : Flutter
- Backend : Spring Boot

6. Appendices

- Detailed database schema
- API documentation
- Future enhancements and AI-based optimizations

This Public Transport Management System SRS is used to design and come up with an effective and intelligent public transportation system.